SOAR Playbook Development

1. Introduction

Many alerts involve repetitive validation steps such as IP reputation checks, ticket creation, and containment actions. Security Orchestration, Automation, and Response (SOAR) platforms reduce manual workload by automating these repetitive tasks.

This report demonstrates the theoretical understanding and practical implementation of a SOAR playbook designed to automatically respond to phishing-related alerts. The implementation integrates SIEM alerts with automated containment and case management.

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2. Theoretical Knowledge

2.1 What is SOAR?

SOAR stands for:

- **Security Orchestration**
- **Automation**
- **Response**

It enables organizations to:

- Integrate multiple security tools.
- Automate repetitive workflows.
- Reduce Mean Time to Respond (MTTR).
- Standardize incident handling procedures.

2.2 Core Components of SOAR

1. Orchestration: Integration of different tools into a unified workflow.
   Example: SIEM → Threat Intelligence → Firewall → Case Management.
2. Automation: Execution of predefined actions without human intervention.
   Example: Automatically blocking a malicious IP.
3. Response: Executing containment or remediation steps.
   Example: Isolating an infected endpoint.

2.3 Importance of SOAR in SOC Operations

Without automation:

- Analysts manually validate every alert.
- High false-positive workload.
- Slow incident response.

With SOAR:

- Faster validation.
- Reduced alert fatigue.
- Consistent incident handling.
- Improved SOC efficiency.

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3. Objective of the Playbook

The objective was to design and test an automated phishing response playbook that:

1. Validates suspicious IP reputation.
2. Blocks malicious IP automatically.
3. Creates an incident case.
4. Documents actions for investigation.

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4. Practical Implementation

4.1 Scenario Overview

A phishing alert was generated from Wazuh indicating suspicious inbound email traffic linked to an external IP address. The SOAR playbook was triggered automatically.

4.2 Playbook Workflow Design

Automated Steps:

1. Receive phishing alert from SIEM.
2. Extract source IP.
3. Check IP reputation.

4. If malicious:
   - Block IP via CrowdSec.
   - Create case in TheHive.
5. Notify SOC analyst.
6. Log action summary.

4.3 Playbook Execution Flow

Step 1: Alert Trigger: Phishing alert received from Wazuh.
Step 2: IP Reputation Check: SOAR platform queried threat intelligence database.
Step 3: Decision Logic: If IP flagged as malicious → Continue automation
Else → Mark as false positive.
Step 4: Automated Containment: IP was blocked via CrowdSec firewall rules.
Step 5: Case Creation: A ticket was automatically created in TheHive with:
Alert details,IP address,Severity level,Automated response summary

4.4 Playbook Test Results

Playbook Step	Status	Notes
Check IP Reputation	Success	IP flagged as malicious
Block IP	Success	192.168.1.102 blocked
Create Case	Success	Ticket #SOC-204 generated
Notify Analyst	Success	Email alert sent

The automation workflow executed successfully without manual intervention.

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5. Verification of Containment

To confirm successful blocking:

- Attempted ping test from malicious IP.
- Connection request failed.
- Firewall logs confirmed block rule active.

Containment was verified successfully.

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6. Playbook Summary

A SOAR playbook was developed to automate phishing alert response. Upon receiving a Wazuh alert, the system validated the source IP reputation. If malicious, the IP was automatically blocked via CrowdSec, and a case was created in TheHive. This automation reduced manual workload and improved response efficiency.

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7. Analysis

The automation process demonstrated:

- Reduced response time.
- Elimination of repetitive analyst tasks.
- Consistent containment procedures.
- Automatic documentation.

Compared to manual handling, response time was significantly reduced and the risk of human error minimized.

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8. Recommendations

- Expand automation to malware detection cases.
- Integrate email sandboxing validation.
- Add automated hash reputation checks.
- Implement automated user isolation for compromised endpoints.
- Periodically review and update playbooks.

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9. Conclusion

This task successfully demonstrated the development and testing of a SOAR playbook for automated phishing response. By integrating SIEM alerts with threat intelligence validation, firewall containment, and case management systems, the workflow improved SOC efficiency and reduced response time. Automation enhances operational maturity and ensures consistent incident response procedures.